

A NOTE ON THE ALGEBRA OF ASSERTIONS IN THE FREELY SLAT-ENRICHED CATEGORY OVER A DISTRIBUTIVE COPY-DISCARD CATEGORY

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This note proposes a categorical semantics for imperative programs with monoidal effects and external nondeterminism. This construction allows us to mix probabilistic behaviour with nondeterministic reasoning while avoiding known issues of combining probability and nondeterminism.

Consider a distributive copy-discard category $\mathcal{C}$ and the freely Slat-enriched category $\hat{\mathcal{C}}$ over it. Explicitly, the objects of $\hat{\mathcal{C}}$ are the objects of $\mathcal{C}$ and morphisms $F: A \rightarrow B$ in $\hat{\mathcal{C}}$ are sets of morphisms $A \rightarrow B$ in $\mathcal{C}$, $F \subseteq \mathcal{C}(A, B)$.

Free enrichment constructions preserve monoidal (co)copy-discard structures and distributivity but not the universal property of (co)products. Then, $\hat{\mathcal{C}}$ is a bimonoidal category where the $\otimes$ monoidal structure is copy-discard, while the $\oplus$ monoidal structure is cocopy-codiscard.

We want to define the validity of program triples $\{P\} \{Q\}$. Intuitively, the outputs of the program $\{$ satisfy the property Q whenever the inputs satisfy the property P . We define a syntax for properties and give them semantics as states in $\hat{\mathcal{C}}$.

Fix a signature

$$\Sigma_S = (\mathcal{C}_S, \text{coar}: \mathcal{C}_S \rightarrow \mathbb{N})$$

for states in $\mathcal{C}$ and a set At of "atomic assertions".

Define assertions and their semantics by induction.

SYNTAX	SEMANTICS
$\top$	$\mathcal{C}(1, X)$
$\perp$	$\emptyset$
$P \wedge Q$	$\llbracket P \rrbracket \cap \llbracket Q \rrbracket$
$\uparrow P$	$\{\lambda \mid \exists t \in \llbracket P \rrbracket \lambda \geq t\}$
$\downarrow P$	$\{\lambda \mid \exists t \in \llbracket P \rrbracket \lambda \leq t\}$
P^c	$\{\lambda \mid \lambda \notin \llbracket P \rrbracket\}$
P_0	$\llbracket P_0 \rrbracket_0$ for $P_0 \in At$
0	$\{0_{1, X}\}$
$\llbracket u_1, \dots, u_N \rrbracket_{\lambda_0}$	$\{\llbracket \llbracket u_1, \dots, u_N \rrbracket_{\lambda_0} \rrbracket\}$ for $\lambda_0 \in \mathcal{C}_S$
$\perp_x P \cdot Q$	$\{\llbracket \perp_x \lambda \cdot t \rrbracket \mid \lambda \in P \wedge t \in Q\}$
$P \vdash_b Q$	$\llbracket P \rrbracket \vdash_b \llbracket Q \rrbracket$
$P(e/x_i)$	$\{\llbracket \lambda(e/x_i) \rrbracket \mid \lambda \in P\}$

example. Consider a Set-monad M and the Slat-enriched category $\hat{Kl}M$. We derive the syntax of outcome assertions and their semantics as the instance of the assertions above in $\hat{Kl}M$. The only difference is $\oplus$: instead of a partial sum, $\oplus$, we have a guarded sum, $\vdash_b$, coming from the distributive copy-discard structure of KlM .

OUTCOME SYNTAX	IMPERATIVE SYNTAX
$\top$	$\top$
$\perp$	$\perp$
$\top \oplus$	0
$\varphi \wedge \psi$	$P \wedge Q$
$\varphi \oplus \psi$	$\uparrow(P \vdash_b Q)$ $\oplus$
$\varphi \Rightarrow \psi$	$(\downarrow(P \wedge Q^c))^c$
P_0	P_0